2026Year 2nd Semester Syllabus

Created Date	2026-08-05 13:45:38	Last-Modified	2026-08-05 13:45:38
Course Title	DEEP NEURAL NETWORKS	Course Code-Section	CAS3201-01
Credit/Time/ Experiment,Lab,Practical Technique Time	3/Tue5,6,Thu4	Department	School of Computing
Time	Tue5,6,Thu4	Location	EngHD504
Exam Date & Time	Midterm exam	Final exam	
Class Language	English	Evaluation Type	Absolute evaluation

Instructor's Profile	Name	Kim Jaehyung	Contact Information	Telephone	
	Department	DEPARTMENT OF ARTIFICIAL INTELLIGENCE		Mail	JAEHYUNGK@YONSEI.AC.KR
	Office			Interview information	

TA's Name & Contact Information	Name		Contact Information	Telephone	

Course Description Brief Introduction of the Course	This course is designed to lecture theories of modern deep learning and give chances to students to have practice. After taking the course, we expect students to be able to implement necessary deep neural network modules for their task of interest.			
Course Goals	1.	Korean	딥러닝에 대한 지식 습득	70%
		English	Gaining knowledge about deep learning	
	2.	Korean	약간의 프로그래밍 실습	20%
		English	A bit of hands-on experience in deep learning programming	
	3.	Korean	활발한 토론 참여	10%
		English	Active discussion	
	4.	Korean		0%
		English		
	5.	Korean		0%
		English		

Core Competencies	The total measurable competencies must be 100%. Each course objective should set the competency as 25%. The core and major competencies should equal at least 50%.		
	논리적사고 50%	문제해결능력 30%	프로그래밍능력 20%
Sub-Competencies/Learning Unit1			
Sub-Competencies/Learning Unit2			
Sub-Competencies/Learning Unit3			
Core Competencies(Liberal Arts)Major competency(	**Must reflect the interrelationship between core competencies (elective courses) and major competencies (major studies).**		
논리적사고			
Sustainable Development Goals			

Average Recommended Amount of Learning per	Average Reading Volume	30	Average amount of writing(Based on A4)	1

Course Methods (%) Total Amount 100	Lecture	Practice Training	Presentation	Dabate	Team Project
	70%	20%	0%	0%	10%

Course Methods 2 Select Relevant Items	PBL Subject	Capstone Design	CBL, Social Innovation Course	Flipped Classroom	Work Experience,Internsh

Grading Policy(%) Total Amount 100 Free Input for Other Information	Midterm exam	Final exam	Quiz	Individual Assignment	Team Assignment	Attendance	Others
	30%	30%	0%	15%	15%	10%	0%

Assignment/ Report, Project Guide	Title of Assignment/Project Name, and Method of Filling Out	Submission Deadline	Type of Submission and Method
	1 (or 2) Coding Assignments		TBD
	Final Project		TBD
Prerequisite		Online Course Address	

Course Material	Course Material Name	Author	Publisher	Publish Year	ISBN
주교재	Deep Learning	Goodfellow, Ian	MIT Press	2017	9780262035613
주교재	Dive into Deep Learning	Quinn, Joanne	Convin Publishers	2019	

Main Learner Precautions	
Attatchment	

Weekly Plan

week	Period	Weekly Topic & Contents	Remarks
1	2026-09-01 2026-09-07	Introduction to the course (DL 1)	(9.1.) Fall semester classes begin (9.3. - 9.7.) Course add and drop period
2	2026-09-08 2026-09-14	Math and ML Basics (DL 2,3,4,5)	
3	2026-09-15 2026-09-21	Deep Feedforward Network (DL 4,5 / D2L 3,4,5 (Practice))	
4	2026-09-22 2026-09-28	Regularization for Deep Learning (DL 7)	(9.24. - 9.25.) Chuseok Holidays 09.24 추석, 09.25 추석, 09.26 추석
5	2026-09-29 2026-10-05	Optimization for Training Deep Models (DL 8)	(10.5.) Alternative Holiday of the National foundation Day 10.03 개천절, 10.05 대체공휴일(개천절)
6	2026-10-06 2026-10-12	Buffer	(10.7.) First third of the semester ends (10.9.) Hangul Proclamation Day 10.09 한글날
7	2026-10-13 2026-10-19	Convolutional Neural Networks (DL 9 / D2L 7,8 / Coding assignment 1)	(10.13. - 10.15.) Course withdrawal period
8	2026-10-20 2026-10-26	Mid-term Exam	(10.20. - 10.26.) Midterm Examinations
9	2026-10-27 2026-11-02	Sequence Modeling (DL 10 / D2L 9, 10)	(10.27. - 10.29.) Application Period for S/U evaluation
10	2026-11-03 2026-11-09	Transformers & Language Model (Slides / D2L 15, 16)	
11	2026-11-10 2026-11-16	Representation Learning -Self Supervised Learning (DL 15)	(11.13.) Second third of the semester ends
12	2026-11-17 2026-11-23	Prompt Engineering & Instruction Tuning (Slides)	
13	2026-11-24 2026-11-30	TBD 1	
14	2026-12-01 2026-12-07	TBD 2	
15	2026-12-08 2026-12-14	Final Project Exhibition	(12.8. - 12.14.) Self-study
16	2026-12-15 2026-12-21	Final Exam	(12.15. - 12.21.) Final Examinations

Attendance Requirement

• Students who are absent for more than one-third of the total class hours will receive an F or NP grade regardless of examination results.
• Courses without midterm and/or final examinations will hold regular classes during the examination periods.

Support for Students with Disabilities(SWDs)

• SWDs can request accommodations related to lectures, assignments, or tests by contacting the course professor at the beginning of semester. However, accommodations may vary depending on the essentiality of lecture and discretion of professors.

[Lecture]
- Visual Impairment: alternative, braille, enlarged reading materials, note-taker
- Physical Impairment: alternative reading materials, access to classroom, note-taker, assigned seat
- Hearing Impairment: note-taker/stenographer, recording lecture
- Intellectual Disability/Autism: note-taker

[Assignments and Test]
- Visual/Physical/Hearing Impairment: (reasonable) extra days for submission, alternative type of assignment, extended test time, alternative type of test, arranging separate test room, and proctors, test ghostwriter
- Intellectual Disability/Autism: (reasonable) extra days for submission, alternative type of assignment

Safety Precautions

• Students required to complete laboratory safety training must finish the online training before the semester begins and submit the certificate at the first class session. Failure to do so may result in restricted class participation.

• Students in physical activity courses should complete appropriate warm-up exercises before class. Those with cardiovascular or chronic respiratory conditions are advised to consult a physician before participating.

Course Administration

• Notices regarding class cancellations, make-up classes, classroom changes, and other course-related matters may be posted on LearnUs. For updates, please check LearnUs regularly.